



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester I - UCSC AY2 [held in September/October 2025]

IS1203 — Foundations of Information Systems

(Two (2) Hours)

Answer ALL questions

0011

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this question paper.
- IV. This paper consists of 4 questions in 12 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 1:

There are different types of information systems and computer-based information systems are implemented to meet different requirements.

- (a) What is the difference between closed systems and open systems? Organize your answer in the following table. Provide at least **three (03)** differences.

[06 Marks]

<u>Closed System</u>	<u>Open System</u>

- (b) Describe how an information system (IS) can support the **management** of a business organization in a specific context. Use the following steps to provide your answer.
- State the selected organization (e.g. Hospital).
 - Explain how an information system can support the management of the organization. You may include a diagram to illustrate your answer.

[06 Marks]

i. Organization:

ii. Description:

Index No:

- (c) Identify **at least two (02)** key points for each of the following categories — inputs, processes, outputs, and benefits — of the information system you described in Question 1(b). Present your answer in the following table.

[08 Marks]

<u>Input</u>	<u>Processes</u>	<u>Output</u>	<u>Benefits</u>

- (d) Software is one of the major components of a computer-based information system. It includes sub-components such as free and open-source software and proprietary software.

State **five (05)** other major components of a computer-based information system and list at least **two (02)** sub-components per each.

[05 Marks]

<u>Major Component-1:</u>
<u>Sub-components:</u>
<u>Major Component-2:</u>
<u>Sub-components:</u>

<u>Major Component-3:</u>
<u>Sub-components:</u>
<u>Major Component-4:</u>
<u>Sub-components:</u>
<u>Major Component-5:</u>
<u>Sub-components:</u>

Question 2:

To design an information system for an organization, first we should understand the organization and its business processes well.

- (a) List six (06) common mediating factors that enhance the relationship between an organization and its information systems.

[03 Marks]

1	4
2	5
3	6

- (b) Describe how an information system (IS) can contribute to a specific value-adding process within an organization. Support your answer with a real-world example and a simple process diagram. Use the following steps to formulate your answer:

- Identify a specific organization.
- Describe one core value-adding business process.
- Explain how an information system supports or enhances that process using a diagram.

[08 Marks]

- (c) Compare Business Process Reengineering (BPR) and Business Process Improvement (BPI). Provide at least **three (03)** comparisons.

[6 Marks]

<u>BPR</u>	<u>BPI</u>
1	1
2	2
3	3

- (d) Describe how functions of a **Human Resource Management (HRM) system** can support each of the **four levels**: Strategic, Management, knowledge and operation, of an organization.

[8 Marks]

<u>Strategic-level functions:</u>
<u>Management-level functions:</u>

Knowledge-level functions:

<u>Operational-level functions:</u>

Question 3:

LakApparel is a growing garment manufacturing company. It operates with over 200 machines, including sewing, embroidery, and cutting equipment. Currently, the factory uses an Executive Support System (ESS) to help top management make strategic decisions. However, most operational data (such as machine performance, maintenance needs, and material usage) is collected manually or through separate systems.

The company is now considering implementing an Enterprise Resource Planning (ERP) system to improve operations. The goal is to integrate all machine and resource data into a single platform, with real-time updates and better support for decision-making at all levels.

- (a) List **three (03)** essential functionalities that an ERP system should have to help LakApparel manage its machines and factory resources more effectively.

[03 Marks]

1
2
3

- (b) For storing and managing the data generated by the ERP system at LakApparel, which type of data storage architecture would be most suitable — a Data Warehouse, or Data Lake?

Explain your choice based on the factory's needs (e.g., real-time data, and analytics). Justify why the other option may be less suitable in this specific case.

[05 Marks]

Most suitable database: Data Warehouse/ Data Lake

Explanation and justification:

- (c) How would an ERP system provide better support for strategic level decision-making compared to the current Executive Support System (ESS)? Provide your answer by comparing the functionalities of ERP and ESS for strategic level decision-making. State at least **two (02)** comparisons.

[08 Marks]

<u>Support for Strategic-Level Decision-Making</u>	
<u>ERP</u>	<u>ESS</u>

(d) If LakApparel needs to monitor its machine conditions and quality of the garment, what kind of hardware should they have for the following requirements?

- i. Collect and preprocess data at the machine level before sending them to the ERP
 - ii. Monitoring machine conditions such as energy consumption, heat and speed.
 - iii. Identify fabric inconsistencies such as color variation, thread pulls, or weave issues.
- Explain the suitability of each hardware type you provided.

[09 Marks]

i. Hardware for collecting and preprocessing data at the machine level:

Explanation:

ii. Hardware for monitoring machine conditions:

Explanation:

iii. Hardware for identifying fabric inconsistencies:

Explanation:

Question 4:

There can be issues with acquiring and using information systems.

- (a) What are the **two (02)** major issues with information technology (IT) **infrastructure investments**?

[02 Marks]

1
2

- (b) List **five (05)** key factors that can be considered for assessing performance of an information system.

[05 Marks]

1.	4.
2.	5.
3.	

- (c) You have been invited by your school to give a speech on issues with computer waste.

List **three (03)** major computer waste issues on which you should base your speech. For each issue, briefly describe how it can be addressed (prevented or mitigated).

[09 Marks]

<u>Issue 1:</u>
<u>How to address:</u>
<u>Issue 2:</u>
<u>How to address:</u>

Index No:

<u>Issue 3:</u>
<u>How to address:</u>

- (d) List and briefly describe **three (03)** major health issues related to prolonged computer usage that IT personnel commonly encounter.

[09 Marks]

<u>Issue 1:</u>
<u>Description:</u>
<u>Issue 1:</u>
<u>Description:</u>
<u>Issue 1:</u>
<u>Description:</u>
